1.Enter destination city, miles travelled to get there, and gallons of gasoline used for any number of trips entered at the keyboard (use ctl+z to stop). Use a function to compute miles per gallon. Display the destination city and miles per gallon for each trip entered. Sum the miles travelled and give a count of the number of trips made. Display these at the end of the program.

Input	Process	Output

Miles per gallon	double milesPerGallon(double miles, double gallons)
	{
	if (gallons <= 0)
	return 0.0;
	return miles / gallons;
	}
		City:
		Miles Per Gallon:
		Total Miles:
		Trip count:

City	while (getline(cin, city))
	{
	cout << "Miles travelled: ";
	if (!(cin >> miles))
	break;
	{
	IF/ELSE NETTED STATEMENT
	}
	cout << "Destination city: ";	

Miles Gallons mpg		
Total miles		
count		

2. Allow the employee to enter last name, job code and hours worked (use ctl+z to stop). Use a function to calculate pay. (Job code L is $25/hr, A is $30/hr and J is $50/hr). Give time and a half for overtime. Display last name and pay for each employee. Sum the pay for each employee as well as count the entries made. After all entries are made, compute and display the average pay and the number of entries made.

Input	Process	Output

Overtime time	double hourlyRate(char jobCode)
	{
	switch (toupper(static_cast<unsigned char>(jobCode)))
	{
	Etc.
	}
	}
		Average pay:
		Number of entries made:

Overtime factor	double calculatePay(char jobCode, double hours) - If/return statement to set over time conditions	

hourlyrate	while (cin >> lastName >> jobCode >> hours)
	{
	if (hourlyRate(jobCode) == 0.0)
	{
	cout << "Invalid job code. Use L, A, or J.\n\n";
	}
	else
	{
	pay = calculatePay(jobCode, hours);
	totalPay += pay;
	count++;
	
	cout << "Last Name: " << lastName << "\n";
	cout << "Pay: $" << pay << "\n\n";
	}
	
	cout << "Last name, job code, hours: ";
	}	

Calculate pay		
Lname		
Job code		
Hours Pay Total pay		

count		

3. Allow any number of students to enter their last name and the credits taken (use ctl+z to stop). Charge $250 per credit hour. Use a function to compute total tuition. Display student last name, credits taken and tuition owed. Sum tuition and give a count of the number of students who entered data.

Input	Process	Output

Rate per credit	double computeTuition(int credits)
	{
	return credits * RATE_PER_CREDIT;
	}
		Student Last Name:
		Credits Taken:
		Tuition:
		Number of Students:
		Total Tuition:

Compute tuition	while (cin >> lastName >> credits)
	{
	tuition = computeTuition(credits);
	totalTuition += tuition;
	count++;
	
	cout << "Last Name: " << lastName << "\n";
	cout << "Credits Taken: " << credits << "\n";
	cout << "Tuition Owed: $" << tuition << "\n\n";
	
	cout << "Last name and credits: ";
	}	

LastName		
Credits		
Tuition		
Totaltuition		
Count		

4. Any number of customers will enter a product code (W, C, G) and a quantity (ctl+z to stop). Use a function to determine unit price. Write another function to compute shipping. Then compute the total. Display the product code, unit price, shipping, extended price (quantity x unit price) and total for the order for each entry. Sum and display the total of all entries made.

Product Code	Unit Price	Shipping
W	$10.00	$2.00
C	$15.00	$5.00
G	$20.00	$7.00

Input	Process	Output

Unitprice	Double unitprice(char code)
	Toupper* case insentivity
	{
	Switch condition
	}
	Double shipping (char code)
	Toupper* case insensitive
	{
	Switch condition
	}
		Product Code:
		Unit Price:
		Shipping:
		Extended Price:
		Number of Entries:
		Grand Total:

Shipping		

Char	cout << "Product code and quantity: ";
	while (cin >> code >> qty)
	{
	Etc.
	}	

Qty		
Price Shipping Extended Order total		
Grandtotal		
Extended		

5. Allow students to enter the department and course code as noted below for any number of courses (ctl+z to stop). Use a function to determine the lab fee also in the table below. For each entry display the department, course code and lab fee. Give the total of all lab fees to collect. Compute and display the average lab fee.

Department	Course Code	Lab Fee
CIS	101	$50.00
CIS	121	$100.00
MAT	111	$25.00
MAT	112	$35.00
ENG	100	$55.00
All Other		$50.00

Input	Process	Output

Dept	// Returns the lab fee for a given department and course code
	double labFee(string dept, int code)
	{
	// make department comparison case-insensitive
	for (char &c : dept)
	c = toupper(static_cast<unsigned char>(c));
	
	if (dept == "CIS" && code == 101) return 50.00;
	if (dept == "CIS" && code == 121) return 100.00;
	if (dept == "MAT" && code == 111) return 25.00;
	if (dept == "MAT" && code == 112) return 35.00;
	if (dept == "ENG" && code == 100) return 55.00;
	return 50.00; // all others
	}
		Number of Courses:
		
		Lab fee total:
		
		Average lab fee:

code	While (cin >> dept >> code)
	{
	fee = labFee(dept, code);	

	total += fee;
	count++;
	
	cout << "Department: " << dept
	<< " Course Code: " << code
	<< " Lab Fee: $" << fee <<
	"\n\n";
	
	cout << "Department and
	course code: ";
	}	

fee	If/ else statement for displaying results	
Total		
Count		